

ECON345 Homework 4: Working in R Markdown, Code Hygiene, and `case_when()`

Due: September 24th, 11:59 PM

Starting this week, homework is submitted as a single knitted **R Markdown** document, not a separate `.R` script — this is the format your project write-ups use, so we start the habit now. If it helps you think, sketch code in a scratch script first, but only the `.Rmd` and its knitted PDF get turned in.

Comment your code to explain what you're doing — partial credit rewards a reasonable, well-documented attempt, not just a correct final answer.

Part I: Start Your R Markdown Document (6 pts)

Question 1 (1 pt)

Create a new R Markdown file, `week_04.Rmd`, output to **PDF**. Start it with a setup chunk like this one (mark it `include = FALSE` in your `.Rmd` so it doesn't show up in the knitted PDF):

```
knitr::opts_chunk$set(echo = TRUE)

library(tidyverse)
library(AER)

data("CASchools")
```

Question 2 (4 pts)

Last week (HW3, Part II) you used the `CASchools` dataset to build `stu_teach_ratio`, bin it into `str_level` with `case_when()`, and summarize read and math scores by `str_level`. Move that work into `week_04.Rmd`: put the `mutate()/case_when()` step in one labeled code chunk and the `group_by()/summarize()` step in another. If a piece isn't working yet, get it working here — it's the foundation the rest of this assignment builds on.

Question 3 (1 pt)

Below your summary chunk, in **markdown text** (not a code comment), write the 1–2 sentence interpretation HW3 asked for: which `str_level` group has the highest average math score, and which has the least variation in reading scores?

Part II: Clean Up This Code (7 pts)

Now load CPS1985 (`data("CPS1985")`), also from **AER** in a new chunk. The chunk below runs, but it's a mess: cryptic names, no comments, and an “endless sea” of nested `if_else()`. Rewrite it applying today's code-hygiene practices.

```
x <- CPS1985
x$y <- if_else(x$education < 12, "A",
              if_else(x$education == 12, "B",
                      if_else(x$education < 16, "C", "D")))
z <- x %>% group_by(y) %>% summarize(m = mean(wage))
```

Your rewrite should:

Question 4 (2 pts)

Open with a **header comment** (2–3 lines) stating what the chunk does, its input (the data), and its output (the summary table).

Question 5 (2 pts)

Replace the cryptic names (`x`, `y`, `z`, `m`) with **short but descriptive** snake_case names.

Question 6 (2 pts)

Replace the nested `if_else()` with a single `case_when()` that maps education to *meaningful* labels — e.g. "Less than HS", "High school", "Some college", "College+". Make sure your conditions don't overlap and cover every value.

Question 7 (1 pt)

Add a one-line **section comment** above the summary step describing what it computes.

Part III: Conditional Summary Statistics (7 pts)

Now use `case_when()` on a *continuous* variable, still working in `week_04.Rmd`.

Question 8 (3 pts)

`experience` (years of potential work experience) is continuous, so we can't `group_by()` it directly. Using `mutate()` and `case_when()`, create a categorical variable `experience_level` with **three** groups:

- "Early career" — fewer than 10 years
- "Mid career" — at least 10 but fewer than 25 years
- "Late career" — 25 years or more

Save the result to a new data frame `cps`.

Question 9 (2 pts)

Using `group_by()` and `summarize()`, compute the **mean** wage, **median** wage, and the **number of workers** (`n()`) for each `experience_level`. Save the result to a new data frame.

Question 10 (1 pt)

Present that summary table with `knitr::kable()` — round to 2 digits and add a caption.

Question 11 (1 pt)

Below the table, in **markdown text** (not a code comment): which experience group earns the most on average, and does the pattern match what you'd expect?

Submission

Upload two files to Canvas before the deadline:

- your `week_04.Rmd`, and
- the **knitted PDF**.

As always: comment your code, and if you get stuck, show your work. If your `.Rmd` won't knit, remember the fixes from lecture — get each piece working before you move on to the next, and restart R and knit clean if results look stale.